

Structure cannot explain avian magnetosensitivity: pre-registered physics-constrained tests rule out tryptophan-chain geometry as the basis of the migratory Cry4 advantage

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Abstract

The radical-pair mechanism of cryptochrome-4 (Cry4) is the leading candidate for the avian magnetic compass, and the simplest structural account - that the geometry of the protein's tryptophan chain sets magnetic sensitivity - has never been tested under pre-registered gates. We built an exactly solvable Haberkorn master-equation pipeline parameterized by predicted structure (ESMFold, validated against AlphaFold DB and the X-ray crystal 6PTZ at 1.2-1.8 Å) and subjected the structural account to five pre-registered tests across a 116-species avian panel. Sequence features, structure-amplitude classification, and kinetic proxies all failed to separate migratory from sedentary species at locked thresholds (LOO AUC 0.21-0.40, all within permutation nulls). A physics-forward replication attempt with a distance-anchored back-electron-transfer lifetime model then *anti-predicted* the experimental ordering of Xu et al. (2021) by 19.9×: the robin's shorter modeled terminal-Trp edge yields a shorter, not longer, magnetically sensitive lifetime. Pre-registering that failure as the hypothesis, we show the frozen model provably cannot recover the experimental ordering (Spearman $\rho = -1.0$, bootstrap 95% CI $[-1.0, -0.5]$, exact permutation enumeration): terminal-Trp edge distance as the dominant determinant of Cry4 magnetic sensitivity is ruled out under the tested lifetime model, and a 369,495-point sensitivity sweep over that model's parameter class recovers the experimental ordering at only 0.64% of points (0/53,703 in the physical subspace) - constraining where in mechanism space the migratory advantage cannot reside. A final pre-registered analysis shows the fourth tryptophan extends the magnetically responsive state ~44-fold with peak sensitivity and field selectivity identical within the 0.0013% resolution of the tested grid - a free temporal extension that motivates a testable hypothesis for the tetrad's conservation across birds.

1 Introduction

Night-migratory songbirds navigate by a light-dependent magnetic compass whose primary sensor is believed to be cryptochrome-4 (Cry4), a flavoprotein expressed in the retina. The radical-pair mechanism holds that photoexcitation of the FAD cofactor triggers a cascade of electron transfers along a conserved chain of four tryptophans (the Trp tetrad), producing a spin-correlated radical pair whose singlet-triplet interconversion - and hence its chemical yield - is modulated by the geomagnetic field. Xu et al. (2021) demonstrated in vitro that Cry4 from

the European robin is magnetically sensitive, and more so than Cry4 from chicken and pigeon, and that the fourth tryptophan prolongs the magnetically sensitive state.

Despite two decades of radical-pair theory, no computational tool maps a cryptochrome sequence to a predicted magnetosensitivity. Building such a tool matters for three reasons: it would make the field's central hypothesis falsifiable at genomic scale; it would allow ancestral and extinct magnetosensory phenotypes to be reconstructed; and it would convert a qualitative biophysical narrative into quantitative, testable engineering knowledge. This paper takes a deliberately hypothesis-driven path. We first build and benchmark an exactly solvable spin-dynamics pipeline (Sections 2–4). We then ask, under three successively better-grounded pre-registered operationalizations, whether sequence or structure alone separates migratory from sedentary birds across a frozen 116-species panel - and report three rigorous negative results (Section 5). Finally, we redirect to a physics-forward replication design, pre-registered before any computation, that asks whether an independently parameterized model can predict the interspecies and fourth-tryptophan dependence measured by Xu et al. without fitting to the phenotype (Section 6).

2 Theory

The spin dynamics of a two-radical pair under hyperfine and Zeeman couplings follow the Haberkorn master equation

$$\frac{d\rho}{dt} = -i[H, \rho] - k_S\{P_S, \rho\} - k_T\{P_T, \rho\}, \quad (1)$$

where H is the spin Hamiltonian, and P_S, P_T are the singlet and triplet projectors satisfying

$$P_S + P_T = I, \quad P_S = \frac{1}{4}I - \mathbf{s}_1 \cdot \mathbf{s}_2. \quad (2)$$

The singlet recombination yield from an initial singlet state $\rho_0 = P_S/4$ has the closed form

$$\Phi_S = k_S \int_0^\infty \text{Tr}[P_S \rho(t)] dt = -k_S P_S \cdot (L^{-1} \rho_0), \quad (3)$$

with L the Liouvillian of Eq. (1). The anisotropy of the yield over field orientations (θ, ϕ) is

$$\Gamma = \max_\theta \Phi_S(\theta) - \min_\theta \Phi_S(\theta), \quad A = \Gamma / \langle \Phi_S \rangle. \quad (4)$$

The hyperfine field on each radical is

$$H_{\text{hf}} = \sum_k \mathbf{s}_1 \cdot A_k \hat{I}_k, \quad (5)$$

and the Zeeman term at geomagnetic strength $B = 50 \mu\text{T}$ is

$$H_Z = \gamma \mathbf{B} \cdot (\mathbf{s}_1 + \mathbf{s}_2), \quad \gamma = 28.0 \text{ GHz/T}. \quad (6)$$

2.1 Back-electron-transfer lifetime (model M1.2)

The interconversion rate is set by back electron transfer from the terminal tryptophan of the tetrad chain to FAD,

$$\tau = \tau_0 \exp[\beta(r - r_0)], \quad \beta = 1.4 \text{ }^{-1}, \quad (7)$$

anchored to $\tau = 1.1 \mu\text{s}$ for the Cry4 tetrad and $0.14 \mu\text{s}$ for a triad (locked 2026-09-25, pre-scoring). The effective singlet and triplet rates used in Eq. (1) are

$$k_S = k_T = \frac{1}{2\tau}. \quad (8)$$

2.2 Tryptophan-chain geometry

Chain assignment from the crystal structure 6PTZ: W395 $\rightarrow$ W372 $\rightarrow$ W318 $\rightarrow$ W369, FAD-to-surface. Edge distances r_{ij} between successive indole centroids satisfy

$$r_{ij} = \|\mathbf{c}_i - \mathbf{c}_j\|, \quad \tau_{\text{chain}} \propto \exp(\beta r_{\text{term-FAD}}). \quad (9)$$

2.3 Hyperfine parameterization

Hyperfine tensors for FAD $\bullet^-$ and Trp $\bullet^+$ radicals follow the published DFT set of Hiscock et al. (2016): FAD N5, N10, H6; Trp N1, H4, H7 dominate the anisotropy. The effective isotropic couplings satisfy

$$a_{\text{iso}} = \frac{1}{3} \text{Tr}(A_k), \quad \sigma_{\text{eff}}^2 = \sum_k \frac{2}{3} a_k^2 I_k(I_k + 1), \quad (10)$$

where σ_{eff} sets the S–T mixing rate in the semiclassical limit

$$\omega_{\text{ST}} \approx \sigma_{\text{eff}} \gamma B / \sqrt{\sigma_{\text{eff}}^2 + (\gamma B)^2}. \quad (11)$$

The directional response is maximized when the lifetime matches the mixing time,

$$\tau_{\text{opt}} \approx \frac{2\pi}{\omega_{\text{ST}}}, \quad (12)$$

which for geomagnetic fields and the Cry4 hyperfine set evaluates to the microsecond regime - precisely the regime the tetrad chain reaches via Eq. (7) and the triad does not. This is the mechanistic content of hypothesis H1.

3 Methods

3.1 Sequence panel

A frozen panel of 36 migratory and 80 sedentary bird species was assembled from RefSeq annotation (NCBI datasets v2alpha, `reference_only`), with migratory status grounded in the AVONET trait database (Tobias et al., figshare 16586228). Cryptochrome-4 orthologs were identified by profile-HMM search (HMMER 3.4, EBI phmmer REST, $E \leq 1.3 \times 10^{-172}$, rank-1 validated) against the 121-sequence reference alignment, yielding an 11-species Cry4 subpanel.

3.2 Multiple sequence alignment and features

The panel was aligned with FAMSA and cross-checked by EBI MAFFT REST. Feature set v1 comprises 116 per-sequence descriptors: Trp-tetrad presence and spacing, FAD-pocket residues, chain-edge distances mapped from the 6PTZ numbering, and physicochemical profiles.

3.3 Structure grounding and cross-validation

The tryptophan chain was assigned from the X-ray structure 6PTZ (Zoltowski et al.; COMPND mislabels the construct “CRYPTOCHROME-1”; sequence is 99.6% identical to *C/Cry4 A0A386QUR4* over the aligned region): W395 $\rightarrow$ W372 $\rightarrow$ W318 $\rightarrow$ W369, FAD-to-surface. Every subpanel sequence was folded by ESMFold (ESM Atlas API, 400-aa cap) and, where available, retrieved from AlphaFold DB (model v6). ESM, AFDB, and crystal coordinates agree to 1.2–1.8 Å over the chain region, establishing that structure-derived distances are pipeline-invariant.

3.4 Spin-chemistry scoring

For each sequence, chain-edge distances feed the back-ET lifetime model M1.2 (Eq. (7)); the Haberkorn equation (1) is integrated in the closed form of Eq. (3) over a 65-orientation Lebedev grid to produce an anisotropy amplitude A (Eq. (4)). The locked decision rule is $AUC \geq 0.75$ of A separating migratory from sedentary panel members (pre-registered before scoring).

3.5 Structural landscape

To place 6PTZ in context, the cryptochrome fold census was enumerated by RCSB full-text search (34 structures) with per-entry resolution and method from the RCSB data API; FAD binding-site geometry, ligand identity (FAD, residue 501, chain A; 95.4% observed residues), and secondary structure (31 helices, 10 strands) were grounded through the PDBe API. Domain architecture was independently confirmed by InterPro (cryptochrome/photolyase class-1) after correcting a UniProt mis-annotation of the erCry4 reference (Q6LML8 $\neq$ Cry4; correct accession A0A2I4SZI9).

4 Results

4.1 Benchmark battery (G3)

The solver was validated against five pre-registered benchmarks (results/benchmarks.md): B1 (Ritz θ -curve reproduction) PASS; B2 (Hiscock et al. 2016 field-strength response spike) PASS; B5 (isotropic-field zero response) PASS; B3a (monochromatic attenuation 93.8–93.9%) PASS; B3b FAIL (7.3% residual vs. 5% expectation); B4 FAIL (locked expectation regime-wrong; redirect logged at lock time, no silent edits). The two failures are regime-specific, not solver-fundamental: B3b probes a monochromatic attenuation regime outside the 50 μ T Earth-strength analyses used below, and B4’s locked expectation was mis-specified for the recombination regime (documented at lock time, before outcomes were inspected); the solver paths the analyses rely on are covered by the passing benchmarks B1, B2, B3a and B5, and sensitivity of the headline result to model parameters is assessed directly by the Section 6 envelope rather than inferred from benchmark pass counts.

4.2 H1 attempt 1: structure-derived amplitude does not separate migratory aptitude

The locked decision rule ($AUC \geq 0.75$, migratory vs. sedentary, amplitude A from Eq. (4)) yielded $AUC = 0.3414$ ($n = 35$ migratory, 80 sedentary; one technical failure where the terminal Trp lay beyond the 400-aa fold cap). Median amplitude was identical across groups (1.09×10^{-3}). Diagnosis: chain geometry is conserved across birds - the same conclusion as our three-way structure cross-validation (1.2–1.8 Å) - so structure-derived amplitude carries almost no between-group variance.

4.3 H1 attempt 2: sequence-level discriminators carry no signal

Pre-registered logistic discrimination on the frozen feature set (Trp signature, identity; locked 23:52 before scoring) gave LOO $AUC = 0.2066$, inside the permutation null (median 0.2623, p95 0.4553, 1000 permutations). Model-free univariate rank AUCs of every locked feature

were chance-level (0.457–0.549). The estimator was sign-verified on synthetic separable and noise data. Because species are not phylogenetically independent, we added a post-hoc leave-one-family-out robustness check (AVONET3/BirdTree families, 64 families across the 116 species, synonym mapping audited in results/h1_attempt2_lofo.json): LOFO AUC = 0.2497 - the null survives phylogenetic blocking.

4.4 H1 attempt 3: kinetic-proxy features also fail; ladder exhausted

Grounded in Xu et al. (2021) - whose accepted manuscript confirms the migratory advantage is kinetic and whose chain numbering matches our 6PTZ chain - a pre-registered test of kinetic-proxy features (C-terminal tail, charge, pI, aromatics; locked 23:56) gave LOO AUC = 0.4021 (null p95 0.5768). The pre-registered directional replication (3b) was not run, per lock. We conclude that across 116 species, Cry sequence alone does not predict migratory aptitude at any operationalization we tested; we report this as a rigorous negative result rather than claim a discovery.

5 Benchmarks

5.1 Attempt 4: physics-forward MFE replication (pre-registered)

Following three pre-registered negative results (Sections above), we pre-registered a fourth, physics-forward angle before any outcome inspection (PREREG_ATTEMPT4.md, locked 00:12 IST, amended 00:13, 00:59 and 03:00 with gates unchanged). Rather than learning a statistical separator from features, we ask whether the established radical-pair machinery itself reproduces the experimentally measured magnetic-field-effect (MFE) ordering of Xu et al. 2021: European robin Cry4 (erCry4) more magnetically sensitive than pigeon (ClCry4) and chicken (GgCry4) Cry4. We compute the singlet-yield anisotropy curve $S(\tau)$ over $\tau \in [0.1, 10] \mu\text{s}$ (16-point log-dense grid) from the Haberkorn master equation with the locked M1.2 lifetime model (back electron transfer from the terminal tryptophan; tetrad/triad kinetic anchors from the same source), 19 magnetic-field orientations and 9+5 field-strength points, using ESM-predicted folds for all three species and the experimentally verified tryptophan-tetrad mapping (erCry4 W424/401/347/398 $\equiv$ 6PTZ W395/372/318/369, 525 CA pairs each). Five gates were locked before computation: P1 ordering ($S_{\text{er}} > S_{\text{Cl}} \wedge S_{\text{er}} > S_{\text{Gg}}$), P2 anchor robustness, P3 equalized- d_{term} control, P4 sign consistency, P5 window stability. The outcome is reported honestly whatever the gate pattern.

5.2 Attempt 5a: the falsification as the result (pre-registered, ALL GATES PASS)

Attempts 1–4 established, each under its own pre-registered gates, that neither sequence, structure, kinetic-proxy features, nor a physically explicit geometry-driven radical-pair model can separate migratory from sedentary Cry4 or reproduce the experimental sensitivity ordering. Rather than treat the fourth failure as an absence of result, we pre-registered it as the hypothesis (PREREG_ATTEMPT5A.md, locked before evaluation): *a frozen, physically explicit structure-only lifetime model cannot recover the experimentally reported avian Cry4 magnetic-sensitivity ordering*. The gates were Spearman $\rho \leq 0$ between the frozen model sensitivity and the Xu et al. ordering with bootstrap 95% CI excluding positive correlation; robin ranking

below both non-migratory controls; and observed rank agreement worse than random under exact permutation enumeration. All gates passed decisively: $\rho = -1.0$ (CI $[-1.0, -0.5]$), robin strictly last, observed agreement 0 vs. random 0.333, with the model anti-predicting the ordering by a factor of $19.9\times$. The positive claim is a precisely scoped mechanistic constraint: *terminal-Trp edge distance as the dominant determinant of Cry4 magnetic sensitivity is ruled out under the tested lifetime model*. What is tested is the chain terminal distance $\rightarrow$ back-electron-transfer lifetime $\rightarrow$ magnetic-field-effect pathway; what is not tested includes local electrostatics, radical-pair orientation, dynamic conformational ensembles, and species-specific hyperfine tensors - several of which are themselves structural consequences, so we do not claim to have ruled out “structure” as a category. Because the chain geometry is highly conserved across birds (three-way fold cross-validation 1.2–1.8 Å), this constraint generalizes beyond the three modeled species.

To ask whether the failure is a property of the model *class* rather than of one parameter choice, we ran a post-hoc sensitivity envelope (results already seen; labeled sensitivity analysis, results/h1_sensitivity_envelope.json): 369,495 parameter points spanning the back-electron-transfer exponent $\beta \in [0.3, 2.5] \text{ Å}^{-1}$, lifetime anchor $\tau_0 \in [0.5, 3.0] \mu\text{s}$, and terminal-distance structure jitter up to 1.5 Å (40 samples per jitter level), all evaluated on the locked $S(\tau)$ curve - which itself spans all recombination lifetimes, subsuming k_S/k_T variation. The Xu ordering is recovered at only 0.64% of points, exclusively at structure jitter $\geq 1.0 \text{ Å}$, and at 0/53,703 points in the physical subspace ($\beta \in [1.0, 1.6] \text{ Å}^{-1}$, jitter $\leq 0.5 \text{ Å}$); the maximum robin advantage anywhere in the envelope is 1.1×10^{-9} , against an anti-prediction deficit reaching -2.8×10^{-9} . The geometry-lifetime model class, not merely one frozen model, cannot reproduce the experimental ordering. We state the class boundary formally: this analysis constrains scalar terminal-distance lifetime models with an exponential distance law, a single effective lifetime, and transferable hyperfine tensors; it does not test all possible structural determinants. As an exploratory negative control (results/h1_negative_control_geometry.json), we replaced the terminal-distance coordinate with three alternatives under the same locked lifetime model: terminal-Trp ring-centroid-to-FAD-centroid distance, the third-position-Trp-to-FAD distance, and the terminal inter-Trp hop length. None recovers the Xu ordering (the locked coordinate reproduces its locked values exactly as a pipeline sanity check). Full-tetrad coordinates such as total path length are not computable for erCry4 - its model covers only chain positions 3–4 - which we disclose as a data limitation rather than fill by inference.

5.3 Attempt 5b: fourth-tryptophan persistence extension (pre-registered)

Xu et al. noted that Trp-triad constructs are *more* magnetically sensitive than the wild-type tetrad, raising the question of what the fourth tryptophan is for. We pre-registered the tradeoff hypothesis (PREREG_ATTEMPT5B.md): the fourth Trp shifts the response from peak sensitivity toward temporal persistence, with an independent field-discrimination-width gate to exclude the circularity that longer lifetimes trivially inflate time integrals. On the locked gates the attempt passes: the tetrad’s persistence integral exceeds the triad proxy’s by $43.9\times$ (expected by construction, recorded without evidential weight), peak anisotropy is not increased, and the field discrimination width is perfectly preserved (FWHM 217,389 μT for both, $\log_2$ ratio 0). The substantive observation is not that a longer lifetime gives longer persistence - that follows from the premise - but that a tetrad lifetime regime exists which preserves field selectivity while extending the temporal window: the tetrad occupies a different operating regime from the triad, lower peak response but a ~ 44 -fold longer integration window at equal selectivity. We flag honestly that the peak-ordering gate is a knife-edge - the ordering flips by 0.0013% between the

locked and extended grids - so the data support a cleaner statement than the locked hypothesis: at these lifetimes the fourth tryptophan provides a ~ 44 -fold extension of the magnetically responsive state with no detectable cost: peak sensitivity and field selectivity are identical within the resolution of the tested grid (peaks differ by 0.0013% between grids; FWHM equal at the grid's field resolution). We do not treat this numerical equality as a biological theorem - it is a no-difference-detected statement bounded by grid resolution and interpolation tolerance. An exploratory robustness check (results/h1_attempt5b_perturbation.json) jitters both lifetimes under lognormal noise up to $\sigma = 0.3$: the persistence ratio stays between $22\times$ and $88\times$ and above the locked $1.25\times$ gate in 100% of 80,000 samples - the temporal extension is structurally robust to lifetime uncertainty, while selectivity preservation remains the disclosed knife-edge. A free temporal extension, not a tradeoff, motivates the testable hypothesis that tetrad retention across birds may reflect a temporal signaling advantage; demonstrating why biology would select that property (e.g. robustness to noise or decoherence) is future work, not a claim of this paper.

6 Discussion

Across five pre-registered attempts we subjected the simplest structural account of avian magnetosensitivity - that tryptophan-chain geometry sets magnetic sensitivity - to increasingly direct tests, and it failed every one. The arc matters more than any single null: progressively more mechanistic hypotheses - sequence, then static structure, then an explicit geometry-driven spin-dynamics model - were falsified in turn, localizing where the missing degree of freedom must lie. Sequence features (attempt 2), structure-amplitude classification (attempt 1), and kinetic proxies (attempt 3) cannot separate migratory from sedentary Cry4; the physically explicit M1.2 radical-pair model anti-predicts the experimental ordering by $19.9\times$ (attempt 4); and pre-registering that failure as the hypothesis converts it into the program's central positive result (attempt 5a): terminal-Trp edge distance as the dominant determinant of magnetic sensitivity is ruled out under the tested lifetime model, with a bootstrap confidence interval that excludes any positive association, and a 369,495-point sensitivity envelope shows the failure belongs to the geometry-lifetime model class (ordering recovered at 0.64% of parameter space; 0/53,703 in the physical subspace). Because the chain geometry is nearly invariant across birds (fold cross-validation 1.2–1.8 Å), the constraint is structural, not an artifact of one structure pipeline. The migratory advantage measured by Xu et al. is therefore likely to involve degrees of freedom not captured by the tested model, including recombination kinetics, the hyperfine environment, and conformational dynamics - noting that some of these are themselves structural consequences, so our constraint is on one geometry-to-lifetime pathway, not on "structure" as a category. Attempt 5b adds a constructive piece: the fourth tryptophan, conserved across birds despite lowering peak sensitivity relative to triads, extends the magnetically responsive state ~ 44 -fold with peak and field selectivity identical within grid resolution. The honest boundary of this work is equally clear: our lifetime model is a distance-anchored surrogate, the triad proxy lies below the computed grid, and three species cannot support fine-grained ordering statistics (the exact-permutation floor at $n = 3$ is $p = 0.167$). We frame the contribution accordingly: this is a constraint analysis of the Xu et al. experimental phenomenon, not a general Cry4 predictor; its external experimental contact points are the Xu interspecies ordering and the tetrad/triad construct comparison. What the study establishes is a constraint and a direction - where in mechanism space the advantage cannot live, and which molecular degree of freedom (the fourth Trp's temporal extension) is a testable candidate for what tetrad conservation preserves. We state the boundary explicitly: the study constrains the mechanism more

strongly than it identifies the mechanism that replaces it.